

BIOCHEMISTRY & MOLECULAR BIOLOGY

POST-TRANSLATIONAL MODIFICATION

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CO- AND POSTTRANSLATIONAL Modifications of Proteins

How proteins are processed, modified, activated, and removed

MAKE

MODIFY

FUNCTION

REMOVE

What are co- and posttranslational modifications?

Proteins are sometimes changed while they are being made or after they are made.

These changes can remove a piece of the protein or add a chemical group.

The changes help the protein become active, stable, correctly located, or properly regulated.

Co-translational = happens while the protein is still being made.

Posttranslational = happens after protein synthesis is finished.

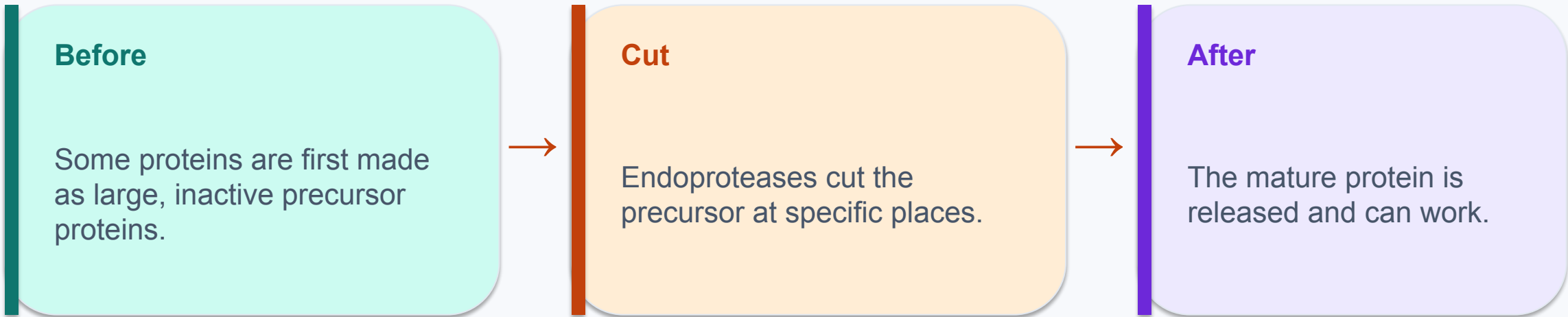


LOCATION IN CELL

Depends on the modification

Ribosome, rough ER, Golgi, cytosol, or after secretion

Trimming: removing an extra piece



Example:

- Insulin is produced as a precursor and then processed into its active form by cleaving the developing secretory vesicle.
- Collagen are cleaved after secretion.

 **LOCATION IN CELL**

Rough ER / Golgi / secretory vesicles / after secretion The exact site depends on the protein

2. COVALENT MODIFICATION

4

I. Phosphorylation: adding a phosphate group

What happens?

A phosphate group is added to an amino acid in the protein.

Which amino acids?

Mostly serine and threonine; less often tyrosine.

Who controls it?

Kinases ADD phosphate.
Phosphatases REMOVE it.

Result: **Protein activity can increase OR decrease.**

LOCATION IN CELL

Cytosol and nucleus; also other cellular compartments

The exact location depends on the protein and kinase/phosphatase

II. Glycosylation: adding carbohydrate chains

N-linked

Sugar(carbohydrate) is attached to the amide nitrogen of asparagine.

Main site: rough ER.

O-linked

Sugar(carbohydrate) is attached to the oxygen on the hydroxyl groups of serine, threonine, or hydroxylysine.

Main site: Golgi.

Why? Helps protein folding, stability, recognition, and proper targeting.

 LOCATION IN CELL

Rough ER → Golgi

N-linked mainly in ER; O-linked mainly in Golgi

III. Hydroxylation: important for collagen

Hydroxylation = adding an –OH (hydroxyl) group.

Target

Proline and lysine residues in collagen are extensively hydroxylated.

Requirement

The enzymes are vitamin C–dependent hydroxylases.

Why important?

Helps collagen form a strong, stable structure.

 LOCATION IN CELL

Rough ER

Collagen hydroxylation occurs during processing in the ER

Carboxylation, biotinylation, lipid attachment & acetylation

Carboxylation

Adds a **carboxyl group** to **glutamate**. Vitamin K–dependent carboxylation is important for blood-clotting proteins.

Biotinylation

Biotin is covalently attached to **lysine** residues of certain carboxylase enzymes.

Lipid attachment

Groups such as **farnesyl** can help anchor proteins to cell membranes.

Acetylation

Many proteins are acetylated at the N-terminus. Histone acetylation can affect gene expression.



Multiple locations

Cytosol, nucleus, ER, or membrane-associated sites depending on the reaction

Protein degradation: tagging proteins for destruction

1 • Ubiquitin tag

A small protein called ubiquitin is attached to a defective, misfolded, or short-lived protein.

2 • Recognition

The ubiquitin tag tells the cell that the protein should be removed.

3 • Proteasome

The proteasome breaks down the tagged protein.

📍 LOCATION IN CELL

Cytosol

Ubiquitination and proteasomal degradation mainly occur in the cytosol

What if proteins misfold in the ER?

ER problem

If a protein cannot fold correctly, it may accumulate in the rough ER.

Cell response

The unfolded protein response increases chaperones and reduces new protein synthesis. Misfolded proteins can be sent for degradation.

Think of it as a cellular quality-control system.

 LOCATION IN CELL

Rough ER → cytosol

ER-associated degradation sends certain misfolded proteins to the cytosol for proteasomal destruction

High-yield summary

Trimming

Cuts away an extra protein piece

ER / Golgi / vesicles / after secretion

Phosphorylation

Adds/removes phosphate to regulate activity

Cytosol / nucleus / other compartments

Glycosylation

Adds carbohydrate chains

Rough ER and Golgi

Hydroxylation

Adds –OH; important for collagen

Rough ER

Other modifications

Carboxyl, biotin, lipid, acetyl groups

Varies by reaction

Ubiquitination

Tags proteins for degradation

Mainly cytosol



THANK YOU